McKinsey DSA Interview Questions (Topic-wise)

Arrays & Strings

• Two Sum

• Three Sum

• Product of Array Except Self

• Maximum Subarray

• Merge Intervals

• Rotate Array

• Longest Substring Without Repeating Characters

• Valid Anagram

• Group Anagrams

• String Compression

Sliding Window

• Maximum Sum Subarray of Size K

• Longest Repeating Character Replacement

• Minimum Window Substring

• Permutation in String

• Longest Substring with K Distinct Characters

Hashing

• Top K Frequent Elements

• Subarray Sum Equals K

• Longest Consecutive Sequence

• Contains Duplicate

• First Unique Character

Two Pointers

• Container With Most Water

• Valid Palindrome

• Remove Duplicates from Sorted Array

• Sort Colors

• Trapping Rain Water

Linked List

• Reverse Linked List

• Merge Two Sorted Lists

• Detect Cycle in Linked List
• Remove Nth Node From End
• LRU Cache
• Copy List with Random Pointer

Stack & Queue

• Valid Parentheses
• Min Stack
• Daily Temperatures
• Next Greater Element
• Implement Queue Using Stacks
• Sliding Window Maximum

Binary Search

• Search in Rotated Sorted Array
• Find Peak Element
• First and Last Position in Sorted Array
• Median of Two Sorted Arrays
• Koko Eating Bananas

Trees

• Maximum Depth of Binary Tree
• Diameter of Binary Tree
• Lowest Common Ancestor
• Validate BST
• Binary Tree Level Order Traversal
• Serialize and Deserialize Binary Tree
• Path Sum

BST

• Kth Smallest Element in BST
• Convert Sorted Array to BST
• BST Iterator
• Recover BST

Heap / Priority Queue

• Kth Largest Element
• Merge K Sorted Lists

• Top K Frequent Elements
• Find Median from Data Stream
• Meeting Rooms II

Graphs

• Number of Islands
• Clone Graph
• Course Schedule
• Pacific Atlantic Water Flow
• Graph Traversal (BFS/DFS)
• Shortest Path Problems

Dynamic Programming

• Climbing Stairs
• House Robber
• Coin Change
• Longest Increasing Subsequence
• Longest Common Subsequence
• Word Break
• Knapsack Variants

Backtracking

• Subsets
• Permutations
• Combination Sum
• N-Queens
• Sudoku Solver

Greedy

• Jump Game
• Gas Station
• Task Scheduler
• Non-overlapping Intervals
• Merge Intervals

Trie

• Implement Trie
• Word Search II

• Design Search Autocomplete System

System Design

• URL Shortener
• Ride Sharing System
• Chat Application
• File Storage System
• Notification Service
• Recommendation Engine

SQL & Data Engineering

• Joins
• Window Functions
• Rank/Dense Rank
• Moving Average
• Top N Customers
• Duplicate Records
• Data Warehouse Design
• ETL Pipeline Design